WMCS045-05 2026
Lab Assignment #1
Deadline: 2026-09-09 **Group number: _____**

1. Latency versus Start-up Latency

Explain in your own words the difference between the terms "latency" and "start-up latency".

2. Time Measurements

Explain why cycle count cannot easily be translated into elapsed time. Provide some concrete case where this is difficult or not possible.

3. Matrix-Multiply

a) Implement the following numerical tasks using C or C++:

$$C \leftarrow A \cdot B, \tag{1}$$

where A, B, and C are square matrices of size N. The elements of these matrices are double-precision numbers. Document and return the code as part of your assignment.

b) Measure the latency of the operation as a function of N. You are free to choose the tools, but use more than one tool. Do not forget to document the system, which you use for your experiments.

c) Check for consistency of the results.

d) Translate the measurements into throughput of floating-point operations.

e) Test the impact of compiler optimizations.

f) Try to explain the results and list possible performance limitations.